

Purpose of Developer-ChatGPT Interaction for Resolving Real-World Issues

In this section, we invite you to share your thoughts on how you, as a developer, would use ChatGPT in an issue-tracking system like GitHub Issues.

Your open-ended responses will provide valuable insights beyond structured options.

- * 3. Did you know you can generate a shareable link to a ChatGPT conversation and include it in a GitHub issue or pull request to reference your solution?

Learn more: <https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq>

☐ Yes

☐ No

- * 4. What could be the reasons a developer shares their conversation with ChatGPT in a GitHub issue or pull request?

(Select all that apply)

☐ To suggest a potential approach generated by ChatGPT to initiate discussion

☐ To point directly to a code snippet generated by ChatGPT

☐ To back up one's argument or approach with insights from ChatGPT

☐ To show alternative approaches that ChatGPT suggested

☐ To ensure transparency, part or all of the solution is AI-generated (attribution)

Other (please specify)

- * 5. For each task below, please select all resources you find effective when resolving a real-world issue in GitHub.

(You can select multiple resources per task)

	ChatGPT	Internal/Project Docs	Stack Overflow	Consulting Another Developer	Some Other Resource
Brainstorming/Ideation					
Exploring solutions, comparing tools, or					

generating ideas.

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Examples: "How to implement OAuth in a React app?" or "What architecture fits this feature?"

Code Generation/ Code Search

Writing code snippets, or scripts.

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Examples: "Generate Python API endpoint" or "Search for code to implement a login function"

Validation

Checking correctness, feasibility, or security of solutions.

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Examples: "Is this algorithm efficient?" or "Does this code follow best practices?"

Debugging

Diagnosing errors, fixing crashes, or reproducing bugs.

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Examples: "Why does this API return a 500 error?" or "Resolve a race condition in C++."

Understanding

Reading or explaining unfamiliar code (i.e., logic or functionality).

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Examples: "What does this regex pattern do?" or "<code> What does this code do?"

Refactoring

Improving readability, performance, or maintainability without changing functionality.

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Examples: "Simplify nested loops" or "<code> Make the provided code more efficient"

Please specify the other resource (if checked)

* 6. Rank the following tasks where ChatGPT is most to least effective while resolving an issue.

Here 1 = most effective and 6 = least effective. You can also drag each option to adjust its ranking.

☐	Brainstorming/Ideation
☐	Code Generation
☐	Debugging
☐	Understanding
☐	Validation
☐	Refactoring

* 7. Please briefly explain why you ranked the tasks in that order. What makes ChatGPT especially helpful (or unhelpful) for specific tasks in the issue resolution process?

Short answers are welcome. If nothing comes to mind, you may simply write “No.”

Experience using ChatGPT-Generated Code for Issue Resolution

In this section, we invite you to share your thoughts and experience of using ChatGPT-Generated Code for resolving real-world issues. Your open-ended responses will provide valuable insights beyond structured options.

* 8. How much do you agree or disagree with the following reasons for Not using ChatGPT-generated code when resolving issues?

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Generic Solutions					
<i>ChatGPT provides generic code, not tailored to proprietary/established APIs or systems</i>	☐	☐	☐	☐	☐
Legal/Licensing Risks					
<i>Code may infringe intellectual property or violate project licenses (e.g., GPL, MIT)</i>	☐	☐	☐	☐	☐
Project Code Exposure Risks					
<i>Using ChatGPT might provide the AI access to the proprietary project's code</i>	☐	☐	☐	☐	☐
High maintenance					

effort

*Requires excessive time
to debug/modify
generated code*

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**Fails non-functional
requirements**

*Code generated may not
meet non-functional
requirements, such as
performance and
security standards*

☐☐☐☐☐

**Distraction from root
causes**

*Misunderstand
developer's prompt, very
hard to control it to get
the code that developer
want for issue resolution*

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Skill Degradation

*Relying on ChatGPT for
issue resolution might
diminish core coding
skills and reduce
learning opportunities*

☐☐☐☐☐

**Lack of Evaluation
Mechanisms**

*Lack of tests or
mechanisms to verify
correctness of generated
solutions*

☐☐☐☐☐

**Risk of Hallucinated or
Harmful Code**

*Code generated may
have nonexistent or
unsafe APIs*

☐☐☐☐☐

Please list any other reasons that might stop you from using AI-generated code.

* 9. Can you describe a time when you regretted relying on ChatGPT-generated code to solve an issue in a project? (Source of dissatisfaction)

Short answers are welcome. If nothing comes to mind, you may simply write "No."

* 10. Can you describe a time when you were glad you used ChatGPT-generated code to solve an issue in your project? (Source of satisfaction)

Short answers are welcome. If nothing comes to mind, you may simply write "No."

Domain-Specific Effectiveness of ChatGPT in Issue Resolution

In this section, we invite you to share your thoughts on which issue domains (e.g., backend issues, automation issues) you believe ChatGPT is effective at resolving and why.

* 11. Below is a list of issue domains you may encounter on GitHub. For each domain, please choose:

- One best reason why ChatGPT is least effective in helping resolve issues in that domain
- One best reason why ChatGPT is most effective in helping resolve issues in that domain

If you believe ChatGPT is never helpful in a given domain, select a reason under "Least effective" and set "Most effective" → Not applicable.

If you believe ChatGPT is always helpful in a given domain, select a reason under "Most effective" and set "Least effective" → Not applicable.

Best Reason for **Least Effectiveness** (Select One)

Best Reason for **Most Effectiveness** (Select One)

**Backend
Issues**
(e.g., API
design,

authentication)

Frontend

Issues

(e.g., UI/UX,
React/Vue
logic)

Automation

Issues

(e.g.,
scripting)

Data Analysis

Issues

(e.g., SQL
queries, excel)

DevOps

Issues

(e.g., cloud
infrastructure,
Kubernetes)

**Machine
Learning**

Issues

(e.g., model
tuning,
PyTorch)

Configuration

Management

Issues

(e.g., package
installation,
environment
setup)

Testing

Issues

(e.g.,
JUnit/Jest)

Demographics and Experience

For the remainder of the survey, we will ask questions about your programming experience and

demographics. You can consider all software projects you have worked on while answering questions in this section.

* 12. What is your gender?

- ☐ Man
- ☐ Woman
- ☐ Non-binary / gender diverse
- ☐ Prefer not to say
- ☐ Prefer to self-describe (please specify)

* 13. In what country do you live?

* 14. What is your age?

- ☐ 18-24
- ☐ 25-29
- ☐ 30-34
- ☐ 35-39
- ☐ 40-44
- ☐ 45-49
- ☐ 50 or above 50
- ☐ Prefer not to answer

* 15. How many years of experience of programming do you have? Give your best estimate.

0

30

* 16. In the past year, I have programmed as... (*Check all that apply*)

- ☐ a professional software developer
- ☐ a part of my job, but not as a professional software developer
- ☐ a hobby
- ☐ a student
- ☐ an open-source software contributor
- ☐ Other (please specify)

* 17. Which programming, scripting, and markup languages have you done extensive development work in over the past year?
(*Check all that apply*)

- ☐ JavaScript
- ☐ Python
- ☐ Java
- ☐ C#
- ☐ TypeScript
- ☐ C++
- ☐ Bash/Shell
- ☐ HTML/CSS
- ☐ SQL
- ☐ PHP
- ☐ Other (please specify)